

# Sungjin Bae, Ph.D.

Rehabilitation robotics & neurorehabilitation researcher

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## EDUCATION

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| 2013–2020 | <b>Ph.D., Robotics Engineering</b><br>DGIST, South Korea                         |
| 2011–2013 | <b>M.S., Engineering Science &amp; Bioengineering</b><br>Osaka University, Japan |
| 2007–2011 | <b>B.S., Robotics Engineering</b><br>Ritsumeikan University, Japan               |

## POSITIONS

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|-----------|------------------------------------------------------------------------------------------|
| 2023–now  | <b>Postdoctoral Researcher</b><br>Northwestern University & Shirley Ryan AbilityLab, USA |
| 2021–2023 | <b>Postdoctoral Researcher</b><br>Mechanical Engineering, KAIST, South Korea             |
| 2020–2021 | <b>Postdoctoral Researcher</b><br>Robotics Engineering, DGIST, South Korea               |
| 2015–2019 | <b>Research Assistant</b><br>Robotics Engineering, DGIST, South Korea                    |

## FELLOWSHIPS & HONORS

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|-----------|------------------------------------------------------------------------------|
| 2022–now  | <b>Postdoctoral Fellowship — Ministry of Health and Welfare, South Korea</b> |
| 2022–2023 | <b>Postdoctoral Fellowship — Ministry of Science and ICT, South Korea</b>    |
| 2022      | <b>Best Paper Award — KSME Bioengineering Division</b>                       |
| 2010–2013 | <b>Monbukagakusho (MEXT) Scholarship, full tuition — Japan</b>               |

## SELECTED PUBLICATIONS

- Bae S.J. et al. (2025). Tracking spasticity dynamics following cyproheptadine administration. *Frontiers in Stroke*.
- Bae S.J., Park H.S. (2023). Immersive VR hand rehabilitation with a gesture-controlled rhythm game: an fNIRS study. *IEEE TNSRE*.
- Bae S.J., Lee Y.H., Chang P.H. (2020). Test-retest reliability of brain activation in robotic passive hand movement. *Brain & Behavior*.
- Bae S.J. et al. (2017). Optimal speed for cortical activation of passive wrist movements by a rehab robot. *Front. Hum. Neurosci*.
- Patents: Shoulder Rehabilitation Robot (KR 10-2084168); Hand Rehabilitation Robot (KR 10-1866253).